Distributing Apples

Time limit: 1.00 s **Memory limit:** 512 MB

There are n children and m apples that will be distributed to them. Your task is to count the number of ways this can be done.

For example, if $n=3$ and $m=2$, there are 6 ways: $[0,0,2]$, $[0,1,1]$, $[0,2,0]$, $[1,0,1]$, $[1,1,0]$ and $[2,0,0]$.

Input

The only input line has two integers n and m.

Output

Print the number of ways modulo 10^9+7.

Constraints

- $1 \le n, m \le 10^6$

Example

Input	Output
`3 2`	`6`